

SENTINEL AI 2.0 - ENTERPRISE SECURITY ARCHITECTURE GUIDE

1. EXECUTIVE SUMMARY & CORE PURPOSE

Sentinel AI 2.0 is a state-of-the-art Enterprise Zero-Trust AI Firewall & Security Gateway designed to sit between corporate users and LLMs (Google Gemini 2.5 / Gemma).

Key Purpose:

- Accidental Data Exposure: Prevents employees from leaking confidential PII & code.
- Prompt Injection & Jailbreaks: Scans incoming prompts in real-time at 100Hz, blocking security bypass attempts, system prompt extraction, and malicious code payloads.

2. ACCESS CONTROL & ROLE PERSONAS

- 1) Super Administrator: anlinpunneli@gmail.com / Anlin20#69 (Level 5 Full Access)
- 2) Security Analyst: alex.mercer@sentinel.local / Sentinel123! (SecOps Audit Mode)
- 3) Standard Employee: employee@sentinel.local / Sentinel123! (Standard User)

3. CODE-BASED ATTACK DEFENSE (4-LAYER BARRIER)

Layer 1: AST Payload Scanner

- Intercepts eval(), exec(), <script>, DROP TABLE, and system("rm -rf") signatures prior to LLM submission. Spikes risk score to 0.98 CRITICAL.

Layer 2: Output XSS Sanitization

- Frontend MarkdownRenderer dynamically escapes raw HTML and script tags into harmless plain text.

Layer 3: Parameterized Data Queries

- Backend storage layers enforce strict object parameterization, rendering SQL and NoSQL injections ineffective.

Layer 4: Sandboxed Code Execution

- Generated code snippets render in static, syntax-highlighted blocks strictly for review.

4. EXTERNAL OUTSIDE CYBER ATTACK DEFENSE

Barrier 1: JWT Bearer Token Gatekeeper

- Direct REST API calls to /api/chat missing valid 256-bit signed JWT tokens are rejected with 401.

Barrier 2: Level 5 System Isolation (Lockdown)

- 1-click Super Admin Lockdown mode freezes all non-admin external API endpoints across the server.

Barrier 3: Strict CORS & Origin Control

- Enforces Cross-Origin Resource Sharing boundaries. Unauthorized external website requests are dropped.

Barrier 4: TLS/SSL HTTPS Tunneling

- Public traffic streams through encrypted TLS endpoints with strict JSON request body caps.

5. LIVE ACCESS ENDPOINTS